Data Visualization Pitch

When Health Outpaces Wealth: A Data-Driven Analysis of Global Longevity and Economic Prosperity (1950-2022)

Abbas Mkhzomi

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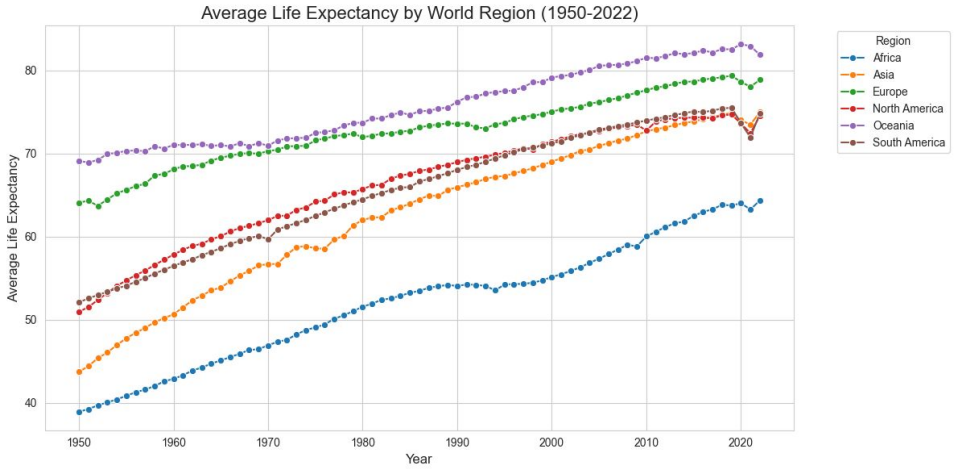
Student Individual Assignment



This analysis examines the evolution of global life expectancy and economic prosperity from **1950** to **2022**, using data from **Our World in Data**. Through a comparative and regional approach, the research focuses on trends across **six major world regions**, highlighting the profound public health transformations, economic shifts, and lingering disparities that have shaped the global population over the past seventy years.

How has the relationship between global economic wealth and life expectancy evolved from 1950 to today?

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Research questions

The analysis aims to provide an overview of the relationship between economic prosperity and population health, relating quantitative trends in **wealth** and **life expectancy** to global development over the past decades.

Some of the research questions considered are:

- How has the relationship between **global economic wealth** (GDP per capita) and **life expectancy** evolved from 1950 to the present day?
- To what extent does a country's economic prosperity directly correlate with the longevity of its citizens?
- How do these health and economic growth trajectories differ across major regions of the world (e.g., Africa, Asia, Europe, the Americas)?
- Are there notable outliers or historical **anomalies** where significant economic wealth did not immediately translate to higher life expectancy, or vice versa?

About Data

Dataset: Life Expectancy vs. GDP per Capita (Our World in Data)

Source: https://ourworldindata.org/grapher/life-expectancy-vs-gdp-per-capita

The dataset, aggregated by **Our World in Data (OWID)**, combines national health and economic metrics from 1543 to the present day. It is shared under a Creative Commons (CC BY 4.0) license. Finding a unified dataset that perfectly aligns economic and health metrics across all years can be challenging, making this **pre-merged** dataset ideal.

Based on robust sources like the UN World Population Prospects and the Maddison Project Database, the data is considered highly reliable. However, limitations exist: **historical data** (especially early decades for developing nations) often relies on estimates or statistical reconstructions.

It is well-structured and easily usable and required only minimal cleaning of the numerical formats.

Methodology

- **Data Collection:** The primary dataset was acquired via manual download from the Our World in Data (OWID) platform in CSV format. A secondary GeoJSON spatial dataset was sourced to provide the geometric boundaries required for the global heatmap.
- **Data Processing & Cleaning:** Used the pandas library within a Jupyter Notebook for initial exploratory data analysis (EDA). Cleaning involved handling **null values**.
- **Data Transformation:**
  - **Scaling:** Applied logarithmic transformations to the GDP per capita axis in both the scatter plot and heatmap to account for exponential wealth disparities.
  - **Merging:** For the heatmap visualization, the OWID dataset was programmatically joined with the GeoJSON features using country codes to map the statistical data to spatial coordinates.
- **Use of AI Tools:** Used Google Gemini as a pair-programming and analytical assistant.
- **Analytical Techniques:**
  - **Descriptive Statistics & Correlation:** Analyzed baseline metrics (means, extremes) and Pearson correlation coefficients.
  - **Outlier Detection:** Used conditional filtering to identify **anomalies** (e.g., countries with high GDP but low life expectancy).
- **Reproducibility:** The static analyses are documented in a shared Jupyter Notebook at Github, and the interactive tools are built with D3.js and hosted publicly at visualization.abbasudo.ir.

Insights from the Data

To extract these insights, I performed exploratory data analysis (**EDA**) using Python (Pandas library). Operations included calculating descriptive statistics (means, minimums, maximums), conducting a **Pearson correlation** analysis to quantify the relationship between the two variables, and applying conditional filtering to identify recent anomalies and outliers.

- There is a moderate positive correlation (**0.6066**) between GDP per capita and life expectancy across the **13,239 records** analyzed. Generally, wealthier populations live longer, but wealth is not the sole determinant of health.
- The dataset highlights extreme historical contexts. The highest life expectancy recorded was 85.3 years (Hong Kong, 2019), while the lowest plummeted to 11.3 years (Cambodia, 1977), reflecting the catastrophic impact of conflict and the **Khmer Rouge regime** rather than purely economic factors.
- As of the most recent data (2022), the global average life expectancy sits at 72.7 years, paired with an average GDP per capita of $19,778.71. **Qatar** holds the historical record for highest wealth at $160,051.23 (2016).

Insights from the Data

- The data reveals compelling deviations from the trend:
  - Wealth without Longevity: Countries like Equatorial Guinea, Russia, and the Seychelles generate **above-average wealth** but suffer from **below-average life expectancy**.
  - Health on a Budget: Conversely, 35 countries (including Albania, Algeria, and Argentina) achieve above-average life expectancy despite having below-average GDP per capita.

Connecting to the Visualizations:

- The **scatter plot** will perfectly illustrate the 0.60 correlation, visualizing the general upward trend while clearly exposing the outliers (like the "Wealth without Longevity" nations) falling far outside the main cluster.
- **Heat Map** will spatially contextualize the 2022 baselines and historical shifts, allowing viewers to see exactly which regions are driving the global averages and where health/wealth disparities are geographically concentrated.

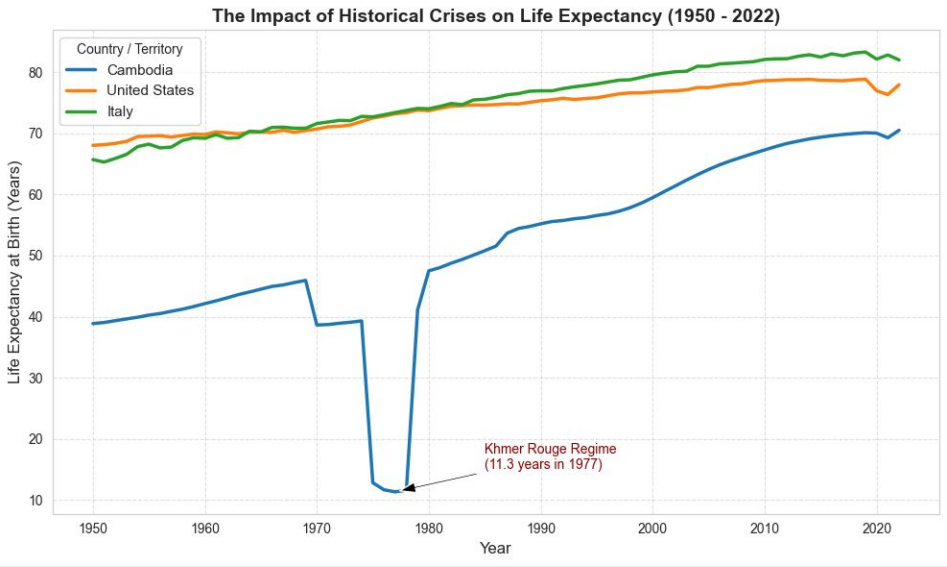
The Fragility of Public Health

The Steady Climb vs. Catastrophic Collapse

While economic development generally pushes life expectancy upward over time—as seen in the consistent growth of the **United States** and the rapid, sustained rise of **Italy** to the global peak (85.3 years)—**progress is not guaranteed.**

The chart highlights a devastating outlier identified in our initial data analysis: Cambodia in **1977.** Life expectancy plummeted to a mere 11.3 years.

This visualization demonstrates that while wealth and GDP (which we will explore next) are strong predictors of health, extreme political instability, conflict, and humanitarian crises can instantly erase decades of public health progress. Health is as much a **political metric** as it is an economic one.



"**The Impact of Historical Crises on Life Expectancy**" by Abbas Mkhzomi, based on data from Our World in Data (© Our World in Data), is licensed under a Creative Commons CC BY 4.0 license. Created by python on Github

Trends in Wealth and Health in Italy

1870s - 1910s - Early Growth and Catastrophe

Slow, steady agrarian growth post-unification. However, **1918** marks the most severe historical anomaly in the dataset: a catastrophic **12.26-year** drop in life expectancy → driven by the devastation of World War I and the Spanish Flu pandemic.

1940s - The Devastation of World War II

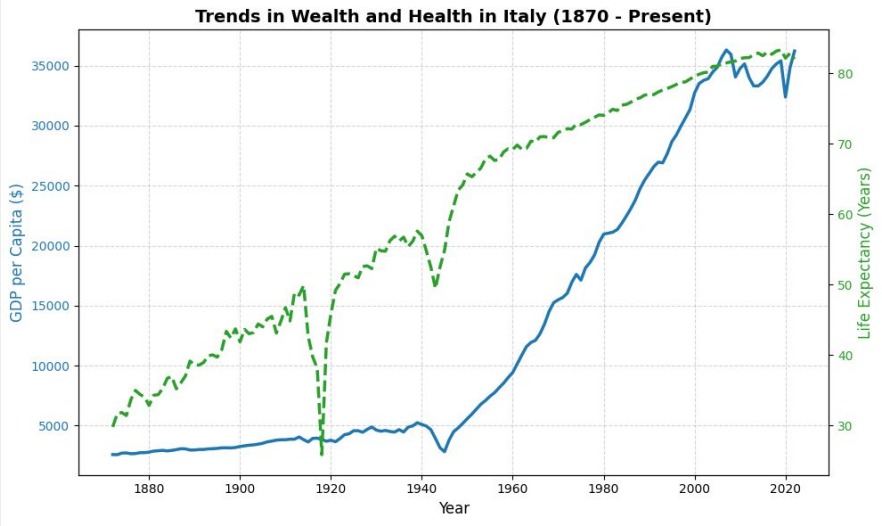
The conflict caused a severe contraction in wealth, with average GDP dropping below **1930s** levels (down to **$4,296**) and a sharp temporary decline in life expectancy → widespread destruction of infrastructure and wartime casualties.

1950s - 60s - The Economic Miracle (Il Boom Economico)

Post-war reconstruction and rapid industrialization led to unprecedented growth. GDP nearly tripled (reaching an average of **$12,199** in the **60s**), and life expectancy leapt to over **70 years** → massive improvements in living conditions, nutrition, and public health.

From 2020 - The COVID-19 Pandemic

As the first major Western epicenter of the virus, Italy suffered a dual shock. The year **2020** saw the largest **single-year** drop in GDP per capita in the dataset (**-$3,021.92**) alongside a noticeable contraction in life expectancy → unprecedented national lockdowns and acute healthcare system strain.



"**Trends in Wealth and Health in Italy (1870 - Present)**" by Abbas Mkhzomi, based on data from Our World in Data (© Our World in Data), is licensed under a Creative Commons CC BY 4.0 license. Created by python on Github.

A 60-Year Transformation

1963 - A World Divided

Looking at the **1963** scatter plot, the global landscape is heavily clustered in the bottom-left quadrant. A massive portion of the world's population lived with a GDP per capita under **$2,000**, and life expectancies in many developing regions struggled to surpass **40-50 years** → Global inequality was stark, with Western nations forming a distinct, isolated wealthy cluster at the top right.

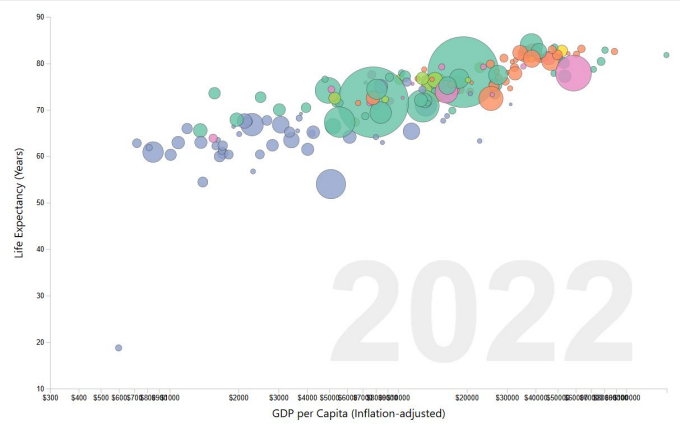
2022 - The Preston Curve in Action

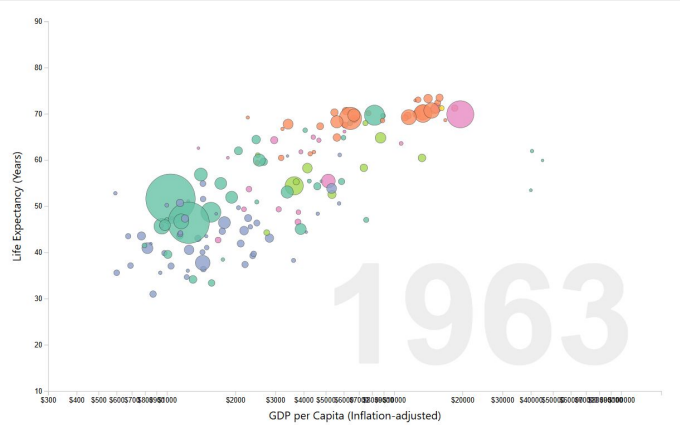
The **2022** chart clearly illustrates the "**Preston Curve**" (the logarithmic relationship between wealth and health).

- **The Steep Climb**: For poorer nations on the left, even small increases in GDP lead to massive, vertical leaps in life expectancy.
- **The Plateau**: For wealthier nations on the right (past **$30k** GDP), the curve flattens. Additional wealth produces diminishing returns for life expectancy.

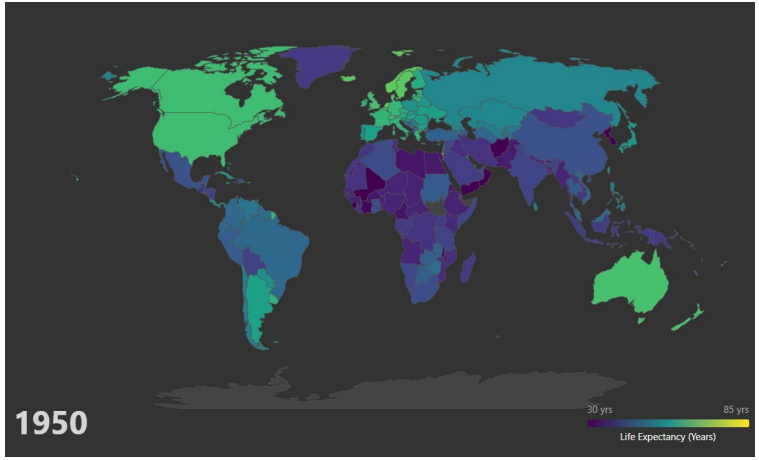
Persistent Inequalities and Outliers

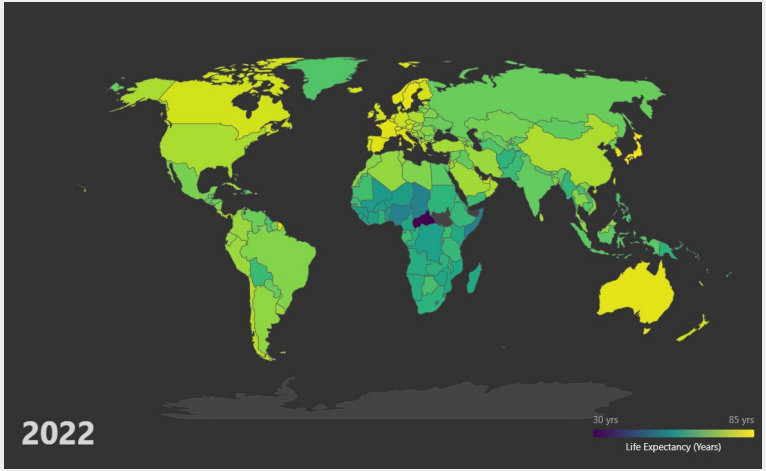
Despite the overarching success, the interactive plot highlights persistent anomalies. As identified in our data analysis, some large bubbles remain notably below the main curve in **2022**. Countries like Equatorial Guinea or Russia demonstrate that high GDP alone does not guarantee long life → Factors like severe inequality, systemic health issues, or conflict can **break the correlation**.

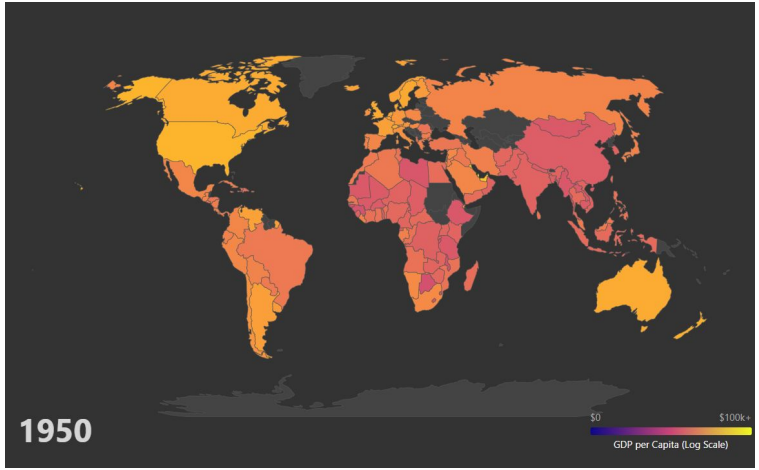


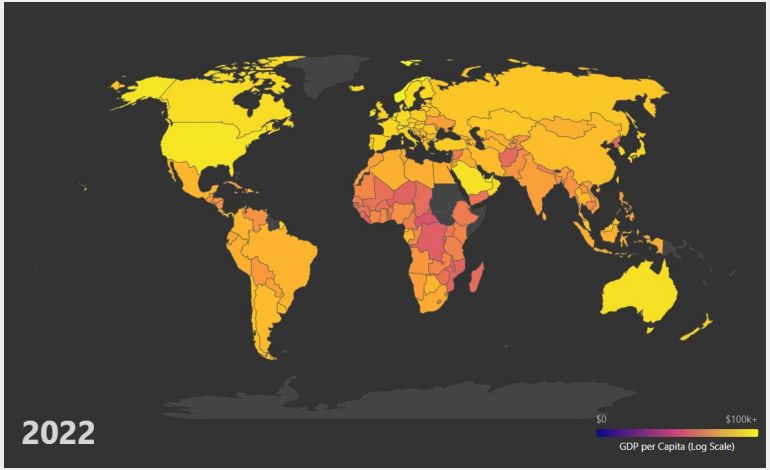


"**Life Expectancy vs. GDP per Capita (1963 & 2022)**" by Abbas Mkhzomi, based on data from Our World in Data (© Our World in Data), is licensed under a Creative Commons CC BY 4.0 license. Created by d3js and hosted at visualization.abbasudo.ir









The Globalization of Health vs. Wealth

"**Global Heat Maps: GDP per Capita vs. Life Expectancy (1950 & 2022)**" by Abbas Mkhzomi, based on data from Our World in Data (© Our World in Data), licensed under CC BY 4.0. Created by d3js and hosted at visualization.abbasudo.ir

The Globalization of Health vs. Wealth

1950: A World of Two Halves

Following **World War II**, global inequality was geographically stark across both metrics. Wealth was heavily concentrated in the "Global North," while vast stretches of Africa, Asia, and South America remained deeply impoverished. The health map mirrored this perfectly: the developing world was dominated by life expectancies frequently below **40 years** (dark blues/purples).

The Health Revolution (Life Expectancy: 1950 vs. 2022)

Looking at the top two maps, the transformation is profound. By **2022**, the globe has overwhelmingly turned green. This visualizes a massive global public health triumph: mass vaccination, sanitation, and antibiotics have universalized basic longevity. Today, a nation no longer needs Western-level wealth to achieve a **70+** year life expectancy.

The Persistent Wealth Divide (GDP: 1950 vs. 2022)

Looking at the bottom two maps, the economic story is much slower. While global GDP has grown, the **2022** map shows that severe economic inequality persists, particularly in Sub-Saharan Africa. The visual contrast between the **2022** Health map and the 2022 Wealth map proves our core insight: Health globalized much faster than wealth. Regional Outliers

Despite the overarching success, the maps highlight localized crises. For instance, in the **2022** Life Expectancy map, the Central African Republic stands out as a dark anomaly in a largely green continent, reflecting how ongoing **conflict** can disrupt even globalized health progress.

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"**When Health Outpaces Wealth: A Data-Driven Analysis of Global Longevity and Economic Prosperity (1950-2022)**" by **Abbas Mkhzomi**.

Developed using charts and analyses produced in Python (Pandas, Matplotlib) and D3.js.

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Data source: Our World in Data (OWID), Life Expectancy vs. GDP per Capita, 2024.

About the Author

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I am a data science master's student with a strong passion for artificial intelligence, data analysis, and web development. I enjoy solving complex problems by blending my experience in backend programming with my growing expertise in data science. My ambition is to build intelligent, data-driven applications that make a real impact.

Core Skills & Expertise

- Data Science & Analysis: Data Mining, Data Visualization, Artificial Intelligence
- Web Scraping: Python, Selenium, BeautifulSoup4
- Backend Development: PHP, JavaScript
- Professional: Teamwork, Collaborative Problem Solving, Project Communication

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